

Visualization Literacy in United States Classrooms: A Qualitative Analysis of State Standards for K–12 Education Policy

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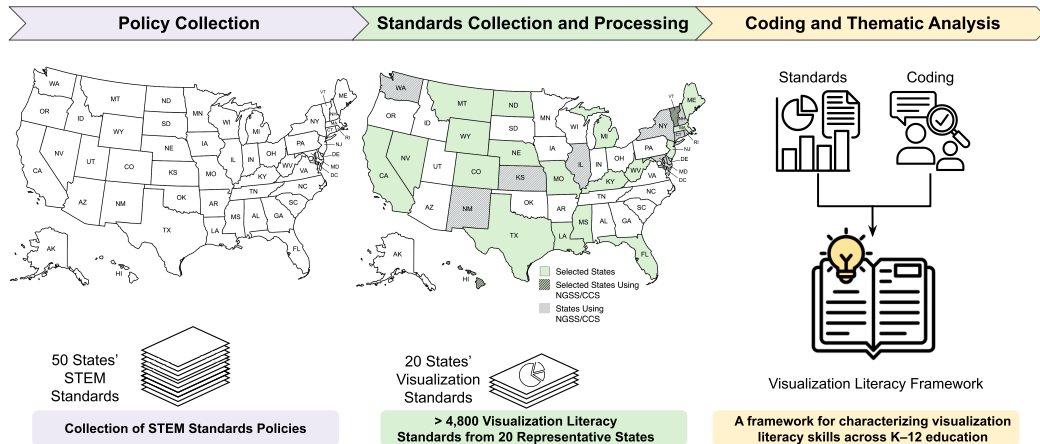


Figure 1: Methodology for constructing the K–12 Visualization Literacy Standards dataset. STEM standards from all 50 U.S. states were collected, 20 representative states selected, and 4,872 standards were coded with visualization literacy metadata.

ABSTRACT

Visualization literacy—the ability to interpret, reason about, and communicate information through visual representations of data—plays a critical role in decision-making, civic participation, and STEM learning. Prior research has primarily examined it through assessment frameworks, educational interventions and materials, and studies of public understanding, but comparatively little is known about how these skills are formally introduced throughout primary and secondary education (K–12) in the United States. State learning standards describe the knowledge and skills students are expected to acquire across grades and subjects, offering a unique perspective into educational expectations surrounding visualization literacy. We compiled STEM education standards from all 50 U.S. states and extracted 4,872 visualization-related standards from a representative subset of 20, each coded by visualization task, idiom, representation mode, grade level, and subject. Preliminary findings show instruction progresses from concrete representations and retrieval-based tasks toward more abstract forms and analytical reasoning. This work contributes a large-scale corpus of policy-derived visualization literacy standards and a foundation for connecting visualization literacy theory with formal educational practice.

Index Terms: Visualization Literacy, K–12 Visualization Standards, Policy Analysis, Qualitative Assessment, Education.

1 INTRODUCTION & RELATED WORKS

Visualization literacy—the ability to interpret visual representations of data and extract meaningful information from them [8]—has become central to visualization research as graphical representations increasingly mediate how individuals access

information and make decisions. These skills influence everyday activities from understanding health information to interpreting financial data and weather forecasts. Existing work has explored visualization literacy through assessment frameworks (e.g., [5, 8]), educational interventions and materials [1, 4], and studies of public understanding [7, 11], but these approaches primarily examine the outcomes of learning and the products of policy rather than the institutional structures that define what students are expected to learn. We instead investigate the policy documents that define educational expectations, a top-down perspective that offers a bridge between visualization literacy theory and educational practice.

In the United States, state governments establish the learning standards that guide curriculum and instruction, providing a unique perspective into how visualization literacy is conceptualized and taught across grades. We conducted a large-scale policy analysis of these standards, collecting documents from all 50 states and extracting 4,872 visualization-related standards from a representative subset of 20, which we coded through qualitative and thematic analysis. This work contributes: (1) a corpus of STEM education standards from all 50 U.S. states as of January 2026; (2) a dataset of visualization literacy standards from 20 representative states; and (3) an analysis of how visualization tasks, idioms, and representation modes vary across grades and subjects.

2 METHODOLOGY

We compiled a corpus of K–12 STEM education standards (mathematics, science, computer science, and technology education) from all 50 U.S. states through a systematic web search. Because most states adapt standards from the Common Core State Standards or Next Generation Science Standards, coding all 50 would have produced largely redundant text. We therefore sampled 20 states by ranking all 50 on five U.S. Census indicators (total population, percent rural, median household income, bachelor’s degree attainment among adults 25 and older, and high school completion among adults 18–24) and selected the two highest-, two median-, and two lowest-ranked states for each (Fig. 1). A standard qualified as a

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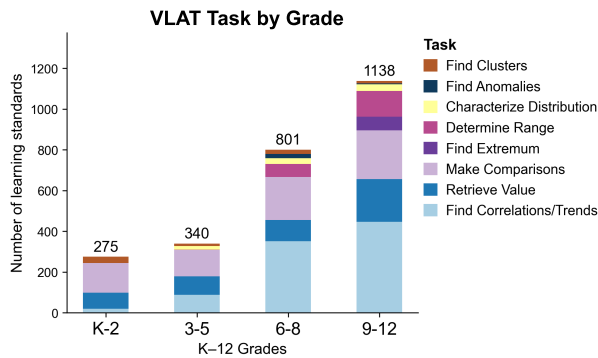


Figure 2: Retrieval and comparison dominate in early grades, while higher-order tasks such as trend identification and range determination become more prominent in grades 6–12.

visualization literacy standard if it referenced a visual representation of data or foundational concepts such as marks and channels. Two researchers independently identified visualization standards, recording grade level, subject area, and context, achieving an average Cohen’s $\kappa > 0.80$ [9]; disagreements were resolved through discussion with a senior visualization expert.

We normalized visualization standards using a semi-automated Python pipeline and grouped using fuzzy string matching [2] into clusters of exact, highly similar, and semantically related standards. Each standard was then coded across four dimensions drawn from established visualization literature: Marks and Channels [10]; Visualization Idioms [3]; Visualization Tasks [6, 8]; and Representation Modes, the medium through which visualizations are encountered, from physical referents to digital displays and XR.

3 RESULTS

Analysis of the visualization literacy standards revealed a consistent progression across K–12 education. First, visualization tasks become increasingly analytical across grades. Not only does the total number of task-coded standards rise across grades, from 275 in grades K–2 to 1,138 in grades 9–12, but the standards also increasingly expect students to *do* more with a visualization beyond producing one (Fig. 2). Value retrieval and comparison are present across all grades, while higher-order tasks demanding greater quantitative reasoning—such as finding correlations and trends, characterizing distributions, and determining ranges—are expected primarily at the upper grade levels. Second, instruction progresses from concrete to abstract representation, from physical referents in early grades toward two-dimensional and digital forms in later grades. Third, the diversity of visualization idioms increases across grades (Fig. 3), growing from 655 idiom-coded standards in grades K–2 to 1,021 in grades 9–12. Tables and diagrams were most common across STEM disciplines and appeared in all grades, while specialized idioms such as distributions and trees and networks were typically introduced in middle and high school. Many standards referenced visual reasoning without specifying a form. These findings suggest that visualization literacy instruction follows a progression from concrete representations and foundational interpretation tasks toward more abstract visual forms and analytical reasoning.

4 CONCLUSION

This work represents an initial effort to connect visualization literacy theory with educational policy. Because standards describe intended rather than enacted instruction, they cannot capture classroom practice. Future work will validate the emerging framework, examine how patterns vary across STEM disciplines and align with existing visualization literacy theories, and conduct educator interviews to understand how standards translate into practice. By exam-

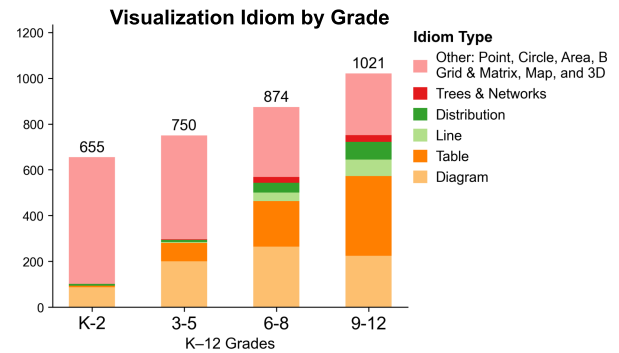


Figure 3: Diagrams appear consistently at all grades, while specialized idioms such as distributions and trees and networks are introduced primarily in grades 6–12.

ining learning standards rather than assessments, interventions, or instructional materials alone, we provide a complementary perspective on how visualization literacy is expected to develop throughout K–12 education.

SUPPLEMENTAL MATERIALS

All standards corpora are available at <https://osf.io/qkwx3>.

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